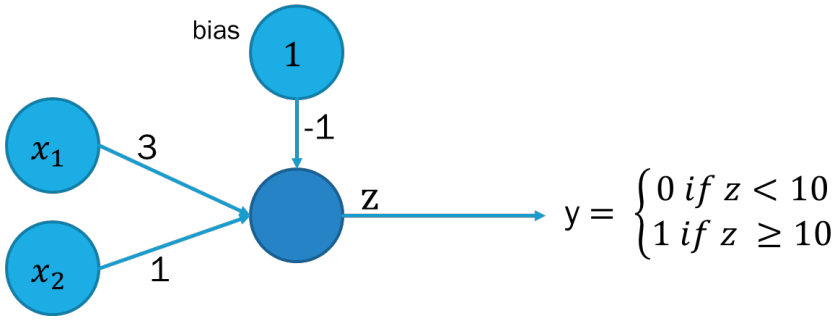
6COSC020W Applied AI

Tutorial Week 8: Artificial Neural Networks

Aim:

- Review the perceptron and forward propagation.
- Experiment with TensorFlow and neural networks.
- Train and evaluate a neural network (model).
- Understand the performance of the network.
- Improve the performance of the network by changing some hyperparameters.

1.- Using forward propagation, calculate the output (y) of the following perceptron with the given inputs:



a) x_1 = 5 and x_2= 10
b) x_1 = 1 and x_2= 2

Compare your results with the person sitting next to you.

2.- Tutorial_Week8.ipynb.

1) Download the file *Tutorial_Week8.ipynb* from Blackboard.
2) Upload and open the file with Google Collaboratory (recommended) or Anaconda (selecting Jupyter Notebook – you may need to install TensorFlow).
3) Work through the Jupyter Notebook. Answer the questions and do the exercises.